Fleet Telematics Lakehouse: Project Proposal

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We will build a pipeline for 5,000 simulated delivery vehicles in 6 days with 3 members. Kafka, Spark, Airflow and Superset run locally in Docker, while storage and queries run on AWS (S3 and Athena).

Data sources

Source	What it gives us	Mode	Lands in
Python generator (simulated)	Telemetry for 5,000 vehicles: GPS, fuel, RPM, engine temp, cargo temp, plus 3 injected anomalies (cold-chain breach, idling, route deviation)	Streaming via Kafka	S3 bronze/telemetry
Reference data (simulated)	Vehicles, drivers, routes, driver-vehicle assignments	Batch CSV	S3 bronze/reference
OSRM routing API (real)	Real road paths for 3 to 5 corridors, saved once as corridors.json	One-off batch	Used by the generator
Open-Meteo weather API (real)	Hourly temperature and rain per corridor, fixed date range	Daily batch (Airflow)	S3 bronze/weather

Flow

1. The generator sends events to Kafka. Malformed events go to a dead-letter topic and an ingestion log.
2. Spark Streaming writes raw events to S3 bronze (append-only). The weather extractor lands its JSON in the same zone.
3. PySpark cleans the data (schema, nulls, duplicates), joins route, driver and weather, runs quality checks, and writes Delta silver tables.
4. PySpark builds the star schema in Delta gold: fact_telemetry, dim_vehicle, dim_driver (SCD Type 2), dim_route, dim_time.
5. Athena queries the gold tables and Superset shows 3 KPIs: fuel efficiency per route, cold-chain compliance %, driver idling scorecard.

6. One Airflow DAG runs steps 2 to 5 in order and keeps the logs.

Local (Docker): Kafka, Spark, Airflow, Superset. AWS: S3, Glue catalog, Athena. If AWS fails, a config switch points storage back to MinIO.

Roles and 6-day timeline

Day	Member 1: generator and ingestion	Member 2: Spark and lakehouse	Member 3: AWS, Airflow, dashboard
1	Event schema, OSRM corridors, 5,000-vehicle reference data	Docker Compose with Spark and Delta, read/write test	AWS account, S3 bucket, IAM user, budget alert, Athena workgroup
2	Generator to Kafka to S3 bronze (10 vehicles)	Silver job: schema, nulls, dedup	Weather extractor to bronze, Airflow skeleton
3	Anomalies, bad events and dead-letter topic, scale to 5,000	Quality checks, join route, driver and weather	Glue and Athena tables, Superset connected
4	Retry with backoff, ingestion logs	Dimensions, fact table, SCD Type 2 MERGE	KPI SQL queries, first dashboard
5	Failure tests, throughput measurements	Anomaly flags checked against injected truth, KPI check in Spark	Full Airflow DAG run, final dashboard
6	Generator and ingestion docs, DLQ evidence	Lakehouse docs, SCD and time-travel evidence	Airflow and dashboard evidence, recorded demo

All three on day 6: README, architecture diagram, slides, rehearsal. Scope is kept small to fit 6 days: 3 to 5 corridors, no Kinesis, EMR or MWAA.